MATH.APP.220 Multivariable Calculus

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Contents (part 5)

1. Double integral
   - Double integral in a rectangular region
   - Double integral in a general region
   - Evaluating a double integral using polar coordinates

The Riemann sum of a single variable function

- Consider a bounded function $f : [a, b] \to \mathbb{R}$. Divide the interval $[a, b]$ into subintervals by the partition points x_i,

$$a = x_0 < x_1 < x_2 < \cdots < x_{n-1} < x_n = b.$$

  The set $P = \{x_0, x_1, \ldots, x_n\}$ is a **partition** of $[a, b]$ and the interval $[x_{i-1}, x_i]$ is the ith **subinterval**, denoted $\triangle x_i$.
- The length of the longest subinterval is called the **norm** $|P|$ of the partition,

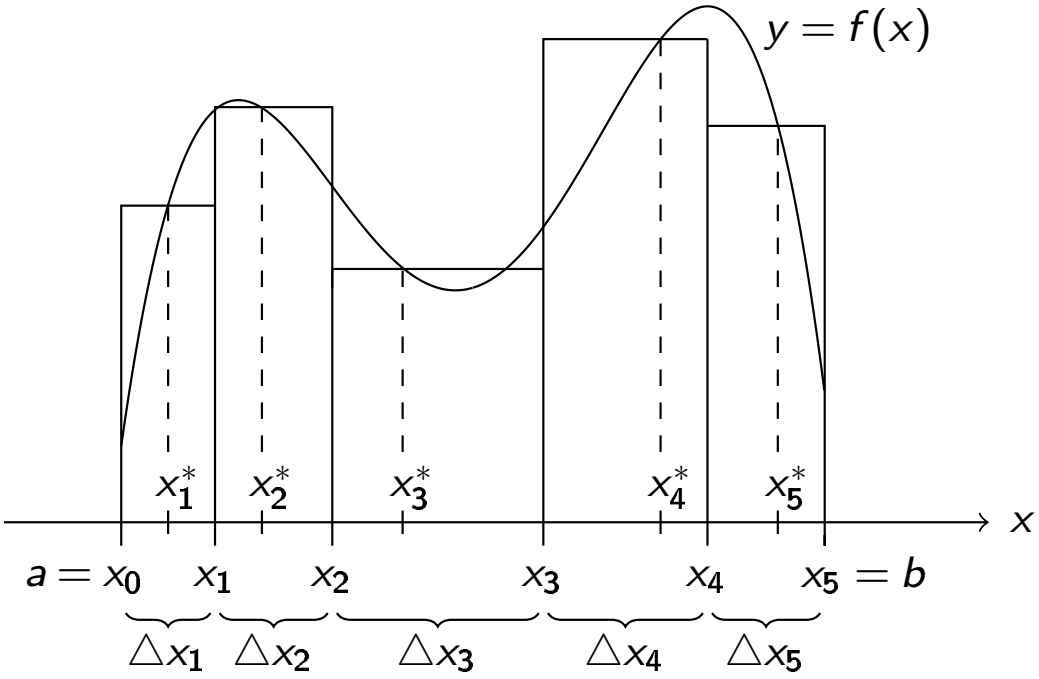
$$|P| = \max\{\Delta x_i : i = 1, 2, \ldots, n\}.$$

- Choose a point x_i^* from each subinterval (then P is called a **tagged partition**), and evaluate $f(x_i^*)$. The Riemann sum is

$$R = \sum_{i=1}^{n} f(x_i^*) \triangle x_i.$$

Graphical representation of the single variable Riemann sum

If $f(x) \geq 0$ for all $x \in [a, b]$, the Riemann sum represents the combined area of rectangles approximating the area between the graph of f and the x-axis.



The integral of a single variable function

- Of course, you can define a Riemann sum also when some or all values of f are negative. In the following definition, f is any function.
- Let $f : [a, b] \to \mathbb{R}$ be bounded. If the limit

$$I = \lim_{|P| \to 0} \sum_{i=1}^{n} f(x_i^*) \triangle x_i = \lim_{n \to \infty} \sum_{i=1}^{n} f(x_i^*) \triangle x_i$$

  exists, f is **integrable** in the interval $[a, b]$ and I is the **(definite) integral** of f over $[a, b]$.
- Then we denote

$$I = \int_a^b f(x)\, dx.$$

Partition of a rectangular region

- Consider now the function $f : R \to \mathbb{R}$, with the rectangular domain $R = [a, b] \times [c, d]$.
- Divide the interval $[a, b]$ in m subintervals using the points x_i, and $[c, d]$ in n subintervals using the points y_j such that

$$a = x_0 < x_1 < x_2 < \cdots < x_{m-1} < x_m = b,$$
$$c = y_0 < y_1 < y_2 < \cdots < y_{n-1} < y_n = d.$$

- These partitions divide the rectangle R in $m \cdot n$ **subrectangles** R_{ij}, where $i = 1, 2, \ldots, m$ and $j = 1, 2, \ldots, n$.
- The set of corner points $P = \left\{ (x_{ij}^*, y_{ij}^*) : i \in \{0, 1, 2, \ldots, m\}, j \in \{0, 1, 2, \ldots, n\} \right\}$ of the subrectangles is the **partition** of $[a, b] \times [c, d]$.

The Riemann sum of a two variable function

- Denote $\triangle x_i = x_i - x_{i-1}$ and $\triangle y_j = y_j - y_{j-1}$. Then the area of the subrectangle R_{ij} is $\triangle x_i \triangle y_j$.
- The **diameter** of the subrectangle is

$$\operatorname{diam}(R_{ij}) = \sqrt{(\triangle x_i)^2 + (\triangle y_j)^2}.$$

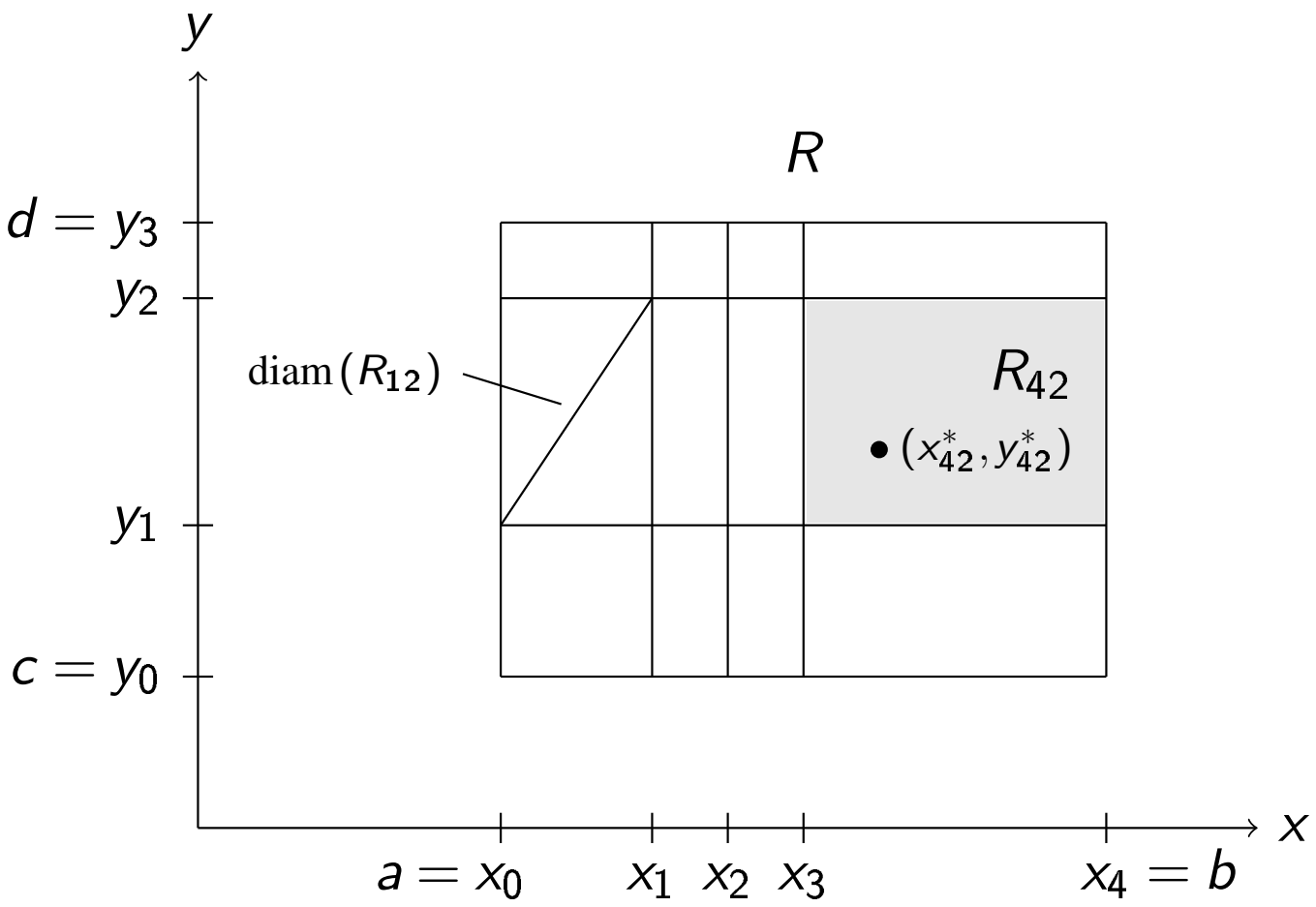
- The **norm** $|P|$ is the longest diameter of the subrectangles

$$|P| = \max\{\operatorname{diam}(R_{ij}) : i = \{1, 2, \ldots, m\}, j = \{1, 2, \ldots, n\}\}.$$

- Choose from each subrectangle R_{ij} a point (x_{ij}^*, y_{ij}^*) (a tag). The **Riemann sum** associated to P and the tags is

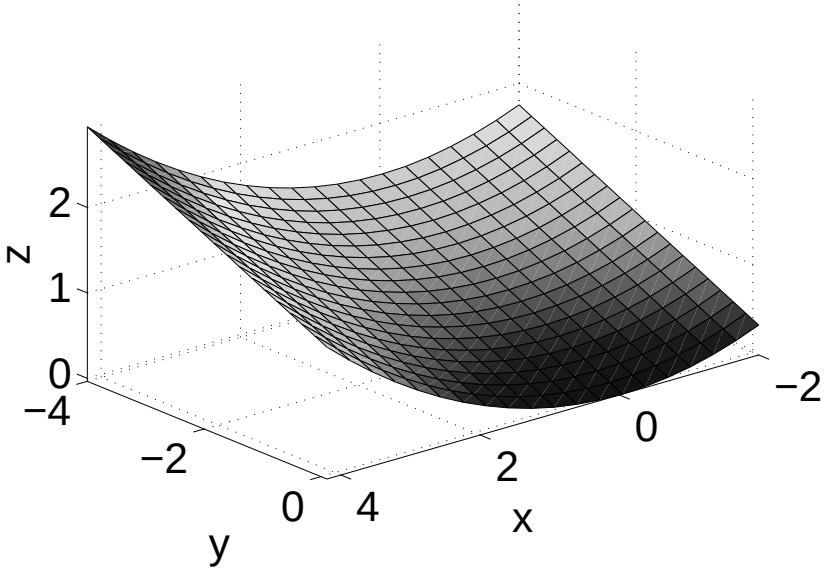
$$\sum_{i=1}^{m} \sum_{j=1}^{n} f\left(x_{ij}^*, y_{ij}^*\right) \Delta x_i \Delta y_j.$$

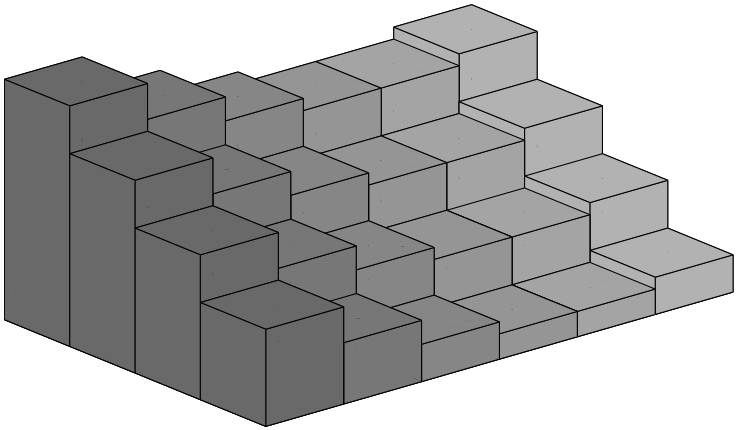
Graphical representation of the partition of a rectangle



Graphical representation or a two variable Riemann sum

If $f(x,y) \geq 0$ for all $(x,y) \in R$, the Riemann sum represents the combined volume of rectangular cuboids approximating the volume between the graph of f and the xy-plane.





Double integral in a rectangular set

- The volume approximation given by the Riemann sum improves, if we increase the number of subrectangles. Then the norm $|P|$ decreases. If $|P| \to 0$, we get the exact volume between the graph of f and the xy-plane.
- Let $f : R \to \mathbb{R}$ be a bounded function defined in the rectangle $R = [a, b] \times [c, d]$. If the limit

$$I = \lim_{|P| \to 0} \sum_{i=1}^{m} \sum_{j=1}^{n} f\left(x_{ij}^*, y_{ij}^*\right) \Delta x_i \Delta y_j$$

exists, f is **integrable** in R and I is the **integral** of f **over** R. Then denote

$$I = \iint_R f(x, y)\, dxdy = \iint_R f\, dA.$$

Evaluation of a double integral in a rectangular set

- Fubini's theorem: The function $f : R \to \mathbb{R}$ that is continuous in the rectangle $R = [a, b] \times [c, d]$, is integrable and

$$\iint_R f(x, y)\, dxdy = \int_a^b \int_c^d f(x, y)\, dydx = \int_c^d \int_a^b f(x, y)\, dxdy.$$

- When evaluating the inner integral with respect to one variable, the other variable is considered a constant.
- You call tell the integrating region is a rectangle if the limits of integration are all constant. If at least one of the limits contains a variable, the region is not rectangular and in that case changing the order of integration means more than just interchanging the limits of one integral with the other.

Multiplicatively separable integrals

- If the integrand is **multiplicatively separable**, i.e. it can be written as $f(x,y) = g(x)\,h(y)$, and the iterated integrals have constant limits of integration, then the double integral can be calculated as a product of separate integrals

$$\int_a^b \int_c^d f(x,y)\,dydx = \left(\int_a^b g(x)\,dx\right)\left(\int_c^d h(y)\,dy\right).$$

- Note that though the integrand is separable, but the integration region is not a rectangle, the double integral cannot be calculated as separate integrals.

Defining a double integral in a general region

- If the function $f : A \to \mathbb{R}$ has a bounded domain $A \subset \mathbb{R}^2$, we can always define a rectangle $R = [a, b] \times [c, d]$ containing the set A.
- Extending f by zero into the set R means defining $f^0 : R \to \mathbb{R}$,

$$f^0(x, y) = \begin{cases} f(x, y), & \text{jos } (x, y) \in A \\ 0, & \text{jos } (x, y) \in R \setminus A \end{cases}$$

- The function f is **integrable** in A , if f^0 is integrable in the rectangle R and the **integral** of f is

$$\iint_A f(x, y)\, dxdy = \iint_R f^0(x, y)\, dxdy.$$

Properties of the double integral

- If the boundary of a closed and bounded set $A \subset \mathbb{R}^2$ consists of finitely many piecewise smooth curves and $f : A \to \mathbb{R}$ is a continuous function, then f is integrable in A.
- The following hold for integrable functions f and $g : A \to \mathbb{R}$ and the value $c \in \mathbb{R}$.
  - $\iint_A cf(x,y)\,dxdy = c\iint_A f(x,y)\,dxdy$,
  - $\iint_A (f+g)\,dxdy = \iint_A f(x,y)\,dxdy + \iint_A g(x,y)\,dxdy$.
  - If $f(x,y) \leq g(x,y)$ for all $(x,y) \in A$, then

$$\iint_A f(x,y)\,dxdy \leq \iint_A g(x,y)\,dxdy.$$

Properties of the double integral

- The following hold for integrable functions f and $g : A \to \mathbb{R}$ and the value $c \in \mathbb{R}$.
  - $\left| \iint_A f(x,y)\, dxdy \right| \leq \iint_A |f(x,y)|\, dxdy.$
  - If A is divided in separate parts A_1 and A_2 (i.e. $A_1 \cup A_2 = A$ and $A_1 \cap A_2 = \emptyset$), then f is integrable in A_1 and A_2, and

$$\iint_A f(x,y)\, dxdy = \iint_{A_1} f(x,y)\, dxdy + \iint_{A_2} f(x,y)\, dxdy.$$

- If the boundary of A consists of finitely many piecewise smooth curves, then the integral of the boundary ∂A is zero and

$$\iint_A f(x,y)\, dxdy = \iint_{\mathrm{int}(A)} f(x,y)\, dxdy = \iint_{\bar{A}} f(x,y)\, dxdy.$$

Vertically simple and horizontally simple sets

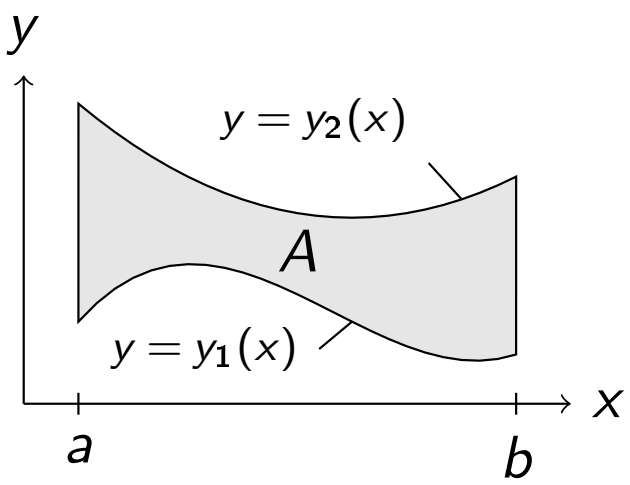
- $A \subset \mathbb{R}^2$ is **vertically simple**, if it has the form

$$A = \{(x, y) : x \in [a, b], y_1(x) \leq y \leq y_2(x)\},$$

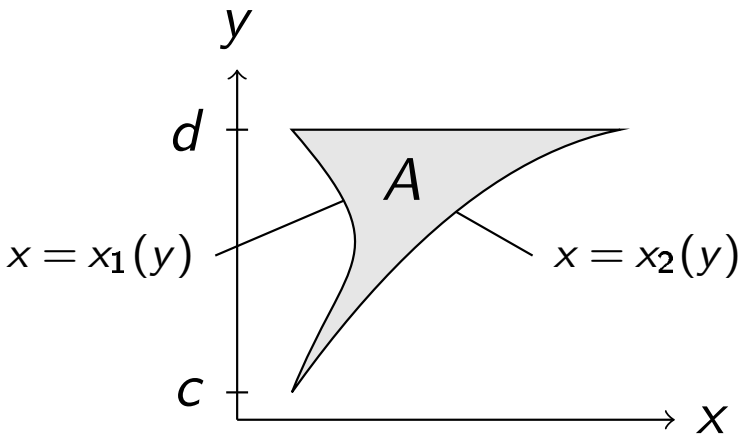
where y_1, y_2 are continuous. A is **horizontally simple**, if

$$A = \{(x, y) : x_1(y) \leq x \leq x_2(y), y \in [c, d]\},$$

where x_1, x_2 are continuous.



vertically simple



horizontally simple

Iterated integrals

- Let $f : A \to \mathbb{R}$ be continuous. If A is vertically simple, then f is integrable and

$$\iint_A f(x,y)\,dxdy = \int_a^b \left(\int_{y_1(x)}^{y_2(x)} f(x,y)\,dy \right) dx,$$

and if A is horizontally simple, then f is integrable and

$$\iint_A f(x,y)\,dxdy = \int_c^d \left(\int_{x_1(y)}^{x_2(y)} f(x,y)\,dx \right) dy.$$

- The outer integral always has constant limits of integration!
- The inner limits of integration can only depend on the integration variable of the outer integral.

Area

- The set A has the **area** $a(A)$, if the constant function 1 is integrable in A. Then

$$a(A) = \iint_A 1\, dxdy.$$

- An example of a set with no area:

$$A = \left\{(x,y) \in \mathbb{R}^2 : x \in [0,1] \cap \mathbb{Q}, y \in [0,1]\right\}.$$

This set A consists of parallel vertical lines in the square $[0,1] \times [0,1]$. The set of lines is so dense it looks like it fills the whole square, but they are still separate lines with no area.

Volume under a surface and mass of a lamina

- Consider the function $f : A \to \mathbb{R}$, where the set $A \subset \mathbb{R}^2$ has an area. If f is integrable and only gets non-negative values, $f(x, y) \geq 0$, then the **volume** between the graph of f and the xy-plane, denoted as the set $T \subset \mathbb{R}^3$, is

$$v(T) = \iint_A f(x, y)\, dxdy.$$

- Let f and A be similar to the above. If $f(x, y)$ represents the density of a lamina (=an flat object) at $(x, y) \in A$, then the **mass** of the lamina is

$$m(A) = \iint_A f(x, y)\, dxdy.$$

Mean value theorem for double integrals

- The set $A \subset \mathbb{R}^2$ is **connected**, if any two points of A can be connected by a continuous curve contained entirely in A.
- Mean value theorem for double integrals: Let $A \subset \mathbb{R}^2$ be closed, bounded and connected, and $a(A) > 0$. If $f : A \to \mathbb{R}$ is continuous, then there exists $(a, b) \in A$ such that

$$\iint_A f(x, y)\, dxdy = f(a, b)\, a(A).$$

- The **average value** of $f : A \to \mathbb{R}$, where $A \subset \mathbb{R}^2$, is

$$\bar{f} = \frac{1}{a(A)} \iint_A f(x, y)\, dxdy.$$

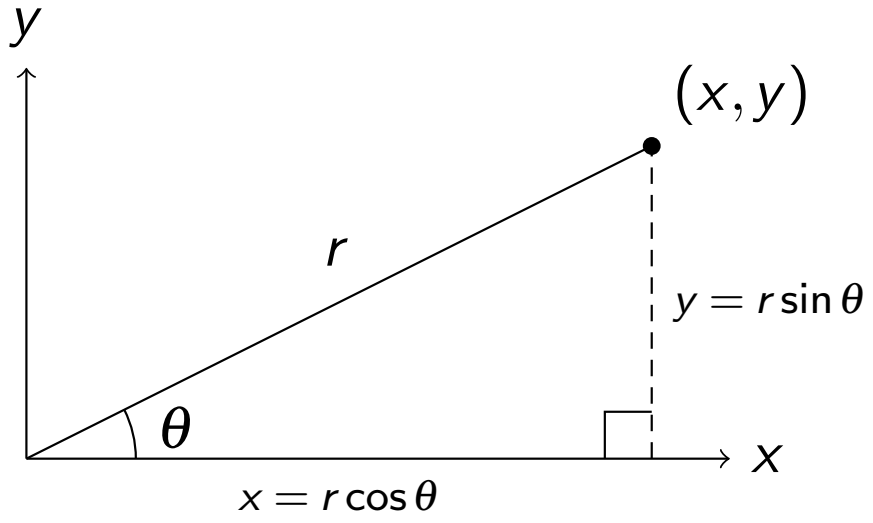
Polar coordinates

- The point $(x, y) \in \mathbb{R}^2$ can be represented by the **polar coordinates** r and θ (=theta).
- The variable r is the distance from the origin (thus $r \geq 0$) and the variable θ is the angle counterclockwise around the origin, the zero angle being on the positive x-axis.
- By the definitions of cosine and sine, $(x, y) = (\cos\theta, \sin\theta)$ is a point on the unit circle and $(x, y) = (r\cos\theta, r\sin\theta)$ is a point on the circle of radius r.
- Therefore we have the coordinate transformation

$$\begin{cases} x = r\cos\theta \\ y = r\sin\theta \end{cases}$$

Polar coordinates

- When $\theta \in \left[0, \frac{\pi}{2}\right]$, we can use the following right triangle.



- The transformation back to cartesian coordinates (usually not used when it comes to integrals) is

$$r = \sqrt{x^2 + y^2},$$
$$\tan\theta = \frac{y}{x}, \quad x \neq 0.$$

Changing into polar coordinates in a double integral

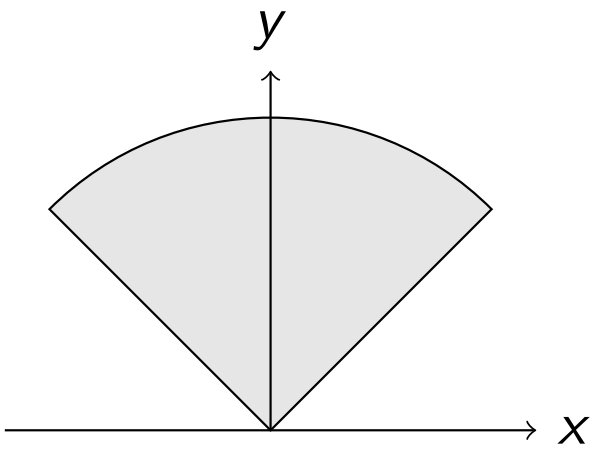
- Let $R \subset \mathbb{R}^2$ be a set containing the points for which, given in polar coordinates, it holds that $\alpha \leq \theta \leq \beta$ and $r_1(\theta) \leq r \leq r_2(\theta)$, where r_1 and r_2 are continuous functions. If $f : R \to \mathbb{R}$ is continuous, then f is integrable and

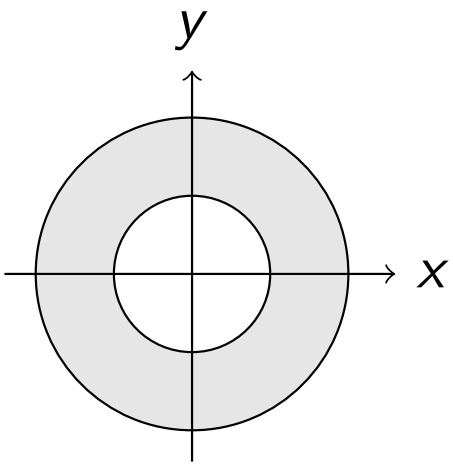
$$\iint_R f(x,y)\,dxdy = \int_\alpha^\beta \int_{r_1(\theta)}^{r_2(\theta)} f(r\cos\theta, r\sin\theta)\,r\,drd\theta.$$

- Note three things about the transformation:
  - change the variables $x = r\cos\theta$, $y = r\sin\theta$ in the function f,
  - multiply the integrand by the Jacobian determinant r,
  - in the limits, r may depend on θ, but not the other way around (often both r and θ have constant limits)

When to change into polar coordinates?

- It is especially beneficial to change into polar coordinates, when the region is a disc or an area between two cocentric circles, or a sector of either one of those, i.e. cut by lines through the origin.





- At least in the case that the circle is centered in the origin, it is recommended to find the limits of r and θ by drawing a figure of the set.